

# AERIS Expert Review Packet

## Anomaly

|                     |                               |
|---------------------|-------------------------------|
| Source / metric     | openaq / pm25                 |
| Observed / expected | 34.5 / 10.75                  |
| Time                | 2026-06-02 14:00 UTC          |
| Location            | 29.8145, -95.3877             |
| Severity            | severe                        |
| Detectors           | zscore, stl, isolation_forest |

## Stored evidence summary

| Source      | Metric             | Unit    | Entities / points | Range; mean                        | Nearest to event                          |
|-------------|--------------------|---------|-------------------|------------------------------------|-------------------------------------------|
| asos        | humidity           | percent | 9 / 623           | 43.99 to 100; mean 81.5836         | 83.7 at 2026-06-02 13:55Z; dt 5 min       |
| asos        | temperature        | degC    | 9 / 623           | 21.6667 to 34; mean 26.8042        | 27 at 2026-06-02 13:55Z; dt 5 min         |
| asos        | wind_direction     | degree  | 9 / 625           | 0 to 360; mean 92.96               | 350 at 2026-06-02 13:55Z; dt 5 min        |
| asos        | wind_speed         | m/s     | 9 / 626           | 0 to 9.77444; mean 2.4596          | 1.54333 at 2026-06-02 13:55Z; dt 5 min    |
| noaa_gfs    | gh_500             | m       | 11 / 132          | 5866.59 to 5901.83; mean 5885.03   | 5880.69 at 2026-06-02 12:00Z; dt 120 min  |
| noaa_gfs    | pbl_height         | m       | 11 / 132          | 14.9006 to 2020.13; mean 544.948   | 47.5406 at 2026-06-02 12:00Z; dt 120 min  |
| noaa_gfs    | precipitable_water | mm      | 11 / 132          | 42.8617 to 54.655; mean 49.4493    | 50.6018 at 2026-06-02 12:00Z; dt 120 min  |
| noaa_gfs    | surface_pressure   | hPa     | 11 / 132          | 1005.44 to 1015.16; mean 1011.19   | 1011.54 at 2026-06-02 12:00Z; dt 120 min  |
| noaa_gfs    | t_850              | degC    | 11 / 132          | 15.9888 to 19.6844; mean 17.7998   | 17.5054 at 2026-06-02 12:00Z; dt 120 min  |
| noaa_gfs    | u_10m              | m/s     | 11 / 132          | -4.09046 to 0.636572; mean -1.6339 | 0.197017 at 2026-06-02 12:00Z; dt 120 min |
| noaa_gfs    | v_10m              | m/s     | 11 / 132          | -3.20242 to 3.45106; mean 0.3664   | -1.07611 at 2026-06-02 12:00Z; dt 120 min |
| openaq      | ozone              | ppm     | 14 / 894          | -0.002 to 0.094; mean 0.0207       | 0.009 at 2026-06-02 14:00Z; dt 0 min      |
| openaq      | pm10               | ug/m3   | 2 / 137           | 9 to 110; mean 25.5839             | 51 at 2026-06-02 14:00Z; dt 0 min         |
| openaq      | pm25               | ug/m3   | 28 / 3022         | -4 to 42.5; mean 5.8391            | 34.5 at 2026-06-02 14:00Z; dt 0 min       |
| openweather | cloud_cover        | percent | 5 / 360           | 0 to 100; mean 55.5056             | 75 at 2026-06-02 13:57Z; dt 2.4 min       |
| openweather | humidity           | percent | 5 / 360           | 57 to 95; mean 81.4833             | 86 at 2026-06-02 13:57Z; dt 2.4 min       |
| openweather | precipitation      | mm/h    | 5 / 360           | 0 to 9.99; mean 0.2948             | 0 at 2026-06-02 13:57Z; dt 2.4 min        |
| openweather | pressure           | hPa     | 5 / 360           | 1012 to 1017; mean 1014.33         | 1015 at 2026-06-02 13:57Z; dt 2.4 min     |
| openweather | temperature        | degC    | 5 / 360           | 22.16 to 34.52; mean 27.3109       | 27.65 at 2026-06-02 13:57Z; dt 2.4 min    |
| openweather | wind_direction     | degree  | 5 / 360           | 0 to 360; mean 110.308             | 340 at 2026-06-02 13:57Z; dt 2.4 min      |
| openweather | wind_speed         | m/s     | 5 / 360           | 0 to 7.2; mean 2.2833              | 1.54 at 2026-06-02 13:57Z; dt 2.4 min     |
| purpleair   | pm25               | ug/m3   | 65 / 4723         | 0 to 58; mean 6.437                | 4.6 at 2026-06-02 14:00Z; dt 0 min        |

| Source     | Metric                       | Unit    | Entities / points | Range; mean                             | Nearest to event                                |
|------------|------------------------------|---------|-------------------|-----------------------------------------|-------------------------------------------------|
| sentinel5p | s5p_aer_ai_granule_available | count   | 6 / 6             | 1 to 1; mean 1                          | 1 at 2026-06-02 19:23Z; dt 323.2 min            |
| sentinel5p | s5p_ch4_granule_available    | count   | 6 / 6             | 1 to 1; mean 1                          | 1 at 2026-06-02 19:23Z; dt 323.2 min            |
| sentinel5p | s5p_co_column                | mol/m^2 | 5 / 5             | 0.028772 to 0.0302015; mean 0.0295      | 0.0288706 at 2026-06-02 19:23Z; dt 323.2 min    |
| sentinel5p | s5p_co_granule_available     | count   | 6 / 6             | 1 to 1; mean 1                          | 1 at 2026-06-02 19:23Z; dt 323.2 min            |
| sentinel5p | s5p_hcho_column              | mol/m^2 | 3 / 3             | 0.000162615 to 0.000169173; mean 0.0002 | 0.000169173 at 2026-06-02 19:52Z; dt 352.2 min  |
| sentinel5p | s5p_hcho_granule_available   | count   | 6 / 6             | 1 to 1; mean 1                          | 1 at 2026-06-02 19:23Z; dt 323.2 min            |
| sentinel5p | s5p_no2_column               | mol/m^2 | 4 / 4             | 3.77616e-05 to 4.38402e-05; mean 0      | 4.09194e-05 at 2026-06-02 19:23Z; dt 323.2 min  |
| sentinel5p | s5p_no2_granule_available    | count   | 6 / 6             | 1 to 1; mean 1                          | 1 at 2026-06-02 19:23Z; dt 323.2 min            |
| sentinel5p | s5p_o3_granule_available     | count   | 6 / 6             | 1 to 1; mean 1                          | 1 at 2026-06-02 19:23Z; dt 323.2 min            |
| sentinel5p | s5p_so2_column               | mol/m^2 | 4 / 4             | -9.16616e-06 to 1.74952e-05; mean 0     | -7.14372e-06 at 2026-06-02 19:23Z; dt 323.2 min |
| sentinel5p | s5p_so2_granule_available    | count   | 6 / 6             | 1 to 1; mean 1                          | 1 at 2026-06-02 19:23Z; dt 323.2 min            |
| tceq       | co                           | ppm     | 3 / 217           | 0 to 0.9; mean 0.2023                   | 0.1 at 2026-06-02 14:00Z; dt 0 min              |
| tceq       | no2                          | ppb     | 13 / 912          | -2.6 to 34.4; mean 6.7047               | 2 at 2026-06-02 14:00Z; dt 0 min                |
| tceq       | so2                          | ppb     | 3 / 215           | -0.6 to 3.6; mean 0.0316                | -0.1 at 2026-06-02 14:00Z; dt 0 min             |

## Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

| Source   | Metric             | Values                                                                                                                                                                  |
|----------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| asos     | humidity           | 06-01 00Z 87.34, 06Z 95.36, 12Z 70.61, 18Z 60.8, 06-02 00Z 84.01, 06Z 94.18, 12Z 76.34, 18Z 66.96, 06-03 00Z 84.58, 06Z 89.65, 12Z 79.25, 18Z 90.35, 06-04 00Z 93.12    |
| asos     | temperature        | 06-01 00Z 25.95, 06Z 24.39, 12Z 29.37, 18Z 31.67, 06-02 00Z 27.27, 06Z 25.14, 12Z 28.78, 18Z 29.95, 06-03 00Z 24.49, 06Z 23.61, 12Z 26.51, 18Z 24.58, 06-04 00Z 24.01   |
| asos     | wind_direction     | 06-01 00Z 127.5, 06Z 21.32, 12Z 80.57, 18Z 135.6, 06-02 00Z 129.4, 06Z 52.16, 12Z 98.15, 18Z 108.9, 06-03 00Z 77.02, 06Z 149.6, 12Z 75.85, 18Z 80.75, 06-04 00Z 63.33   |
| asos     | wind_speed         | 06-01 00Z 1.784, 06Z 0.3106, 12Z 1.281, 18Z 3.782, 06-02 00Z 1.998, 06Z 0.5851, 12Z 1.982, 18Z 3.931, 06-03 00Z 4.389, 06Z 3.077, 12Z 3.296, 18Z 2.883, 06-04 00Z 2.658 |
| noaa_gfs | gh_500             | 06-01 06Z 5900, 12Z 5885, 18Z 5900, 06-02 00Z 5888, 06Z 5898, 12Z 5881, 18Z 5894, 06-03 00Z 5880, 06Z 5880, 12Z 5869, 18Z 5878, 06-04 00Z 5868                          |
| noaa_gfs | pbl_height         | 06-01 06Z 225.8, 12Z 67.97, 18Z 1700, 06-02 00Z 564.7, 06Z 86.98, 12Z 42.25, 18Z 1654, 06-03 00Z 748.6, 06Z 221.8, 12Z 182.3, 18Z 717.2, 06-04 00Z 327.7                |
| noaa_gfs | precipitable_water | 06-01 06Z 46.22, 12Z 46.21, 18Z 48.98, 06-02 00Z 51.86, 06Z 49.79, 12Z 50.16, 18Z 48.67, 06-03 00Z 46.9, 06Z 48.34, 12Z 51.85, 18Z 52.98, 06-04 00Z 51.45               |
| noaa_gfs | surface_pressure   | 06-01 06Z 1011, 12Z 1011, 18Z 1011, 06-02 00Z 1009, 06Z 1011, 12Z 1011, 18Z 1012, 06-03 00Z 1010, 06Z 1012, 12Z 1013, 18Z 1013, 06-04 00Z 1011                          |
| noaa_gfs | t_850              | 06-01 06Z 18.58, 12Z 17.84, 18Z 17.33, 06-02 00Z 18.76, 06Z 18.54, 12Z 17.59, 18Z 16.97, 06-03 00Z 19.2, 06Z 18.95, 12Z 16.66, 18Z 16.32, 06-04 00Z 16.86               |
| noaa_gfs | u_10m              | 06-01 06Z -0.7446, 12Z -0.4489, 18Z -1.458, 06-02 00Z -2.194, 06Z -0.3987, 12Z -0.483, 18Z -1.96, 06-03 00Z -2.718, 06Z -2.461, 12Z -1.27, 18Z -3.141, 06-04 00Z -2.33  |

| Source      | Metric                       | Values                                                                                                                                                                                       |
|-------------|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| noaa_gfs    | v_10m                        | 06-01 06Z 1.967, 12Z 0.7749, 18Z 0.7166, 06-02 00Z 2.65, 06Z 1.596, 12Z -1.005, 18Z -1.215, 06-03 00Z 1.894, 06Z 0.4182, 12Z -2.363, 18Z -0.8762, 06-04 00Z -0.1611                          |
| openaq      | ozone                        | 06-01 06Z 0.009731, 12Z 0.01646, 18Z 0.04762, 06-02 00Z 0.01781, 06Z 0.007731, 12Z 0.01115, 18Z 0.03371, 06-03 00Z 0.02384, 06Z 0.02074, 12Z 0.02067, 18Z 0.02218, 06-04 00Z 0.01369         |
| openaq      | pm10                         | 06-01 06Z 31.42, 12Z 44.58, 18Z 30.83, 06-02 00Z 25.17, 06Z 22.08, 12Z 31.17, 18Z 28.42, 06-03 00Z 17.5, 06Z 15.83, 12Z 22.08, 18Z 15.27, 06-04 00Z 18                                       |
| openaq      | pm25                         | 06-01 06Z 6.451, 12Z 5.868, 18Z 6.815, 06-02 00Z 5.303, 06Z 4.973, 12Z 6.72, 18Z 5.216, 06-03 00Z 5.146, 06Z 4.94, 12Z 7.158, 18Z 5.391, 06-04 00Z 6.993                                     |
| openweather | cloud_cover                  | 06-01 00Z 18.8, 06Z 41.5, 12Z 43.47, 18Z 44.27, 06-02 00Z 20.8, 06Z 24.37, 12Z 54.9, 18Z 83.7, 06-03 00Z 38.03, 06Z 89.53, 12Z 95.43, 18Z 84.5, 06-04 00Z 99.1                               |
| openweather | humidity                     | 06-01 00Z 84.05, 06Z 92.27, 12Z 74.93, 18Z 64.9, 06-02 00Z 80, 06Z 91.3, 12Z 78.7, 18Z 69.03, 06-03 00Z 85.87, 06Z 87.47, 12Z 82.67, 18Z 83.73, 06-04 00Z 92.7                               |
| openweather | precipitation                | 06-01 00Z 0, 06Z 0.005, 12Z 0, 18Z 0.03, 06-02 00Z 0, 06Z 0, 12Z 0, 18Z 1.003, 06-03 00Z 0.2023, 06Z 0, 12Z 0.5023, 18Z 1.794, 06-04 00Z 0                                                   |
| openweather | pressure                     | 06-01 00Z 1013, 06Z 1014, 12Z 1014, 18Z 1012, 06-02 00Z 1013, 06Z 1014, 12Z 1015, 18Z 1013, 06-03 00Z 1015, 06Z 1015, 12Z 1017, 18Z 1015, 06-04 00Z 1015                                     |
| openweather | temperature                  | 06-01 00Z 26.62, 06Z 24.92, 12Z 29.78, 18Z 32.53, 06-02 00Z 28.22, 06Z 25.73, 12Z 29.28, 18Z 31.44, 06-03 00Z 24.26, 06Z 23.55, 12Z 26.61, 18Z 25.7, 06-04 00Z 23.88                         |
| openweather | wind_direction               | 06-01 00Z 153.1, 06Z 79.53, 12Z 122.5, 18Z 133.8, 06-02 00Z 166.2, 06Z 68.2, 12Z 121.1, 18Z 129.9, 06-03 00Z 91.73, 06Z 93.13, 12Z 55.87, 18Z 136.4, 06-04 00Z 69.9                          |
| openweather | wind_speed                   | 06-01 00Z 2.894, 06Z 1.041, 12Z 1.079, 18Z 3.923, 06-02 00Z 3.192, 06Z 0.8987, 12Z 1.722, 18Z 2.616, 06-03 00Z 2.312, 06Z 1.638, 12Z 2.364, 18Z 4.105, 06-04 00Z 1.736                       |
| purpleair   | pm25                         | 06-01 00Z 6.302, 06Z 7.763, 12Z 5.952, 18Z 6.977, 06-02 00Z 5.427, 06Z 5.803, 12Z 6.927, 18Z 4.514, 06-03 00Z 5.967, 06Z 6.665, 12Z 8.411, 18Z 5.723, 06-04 00Z 8.008                        |
| sentinel5p  | s5p_aer_ai_granule_available | 06-01 18Z 1, 06-02 18Z 1, 06-03 18Z 1                                                                                                                                                        |
| sentinel5p  | s5p_ch4_granule_available    | 06-01 18Z 1, 06-02 18Z 1, 06-03 18Z 1                                                                                                                                                        |
| sentinel5p  | s5p_co_column                | 06-01 18Z 0.03012, 06-02 18Z 0.02882, 06-03 18Z 0.02948                                                                                                                                      |
| sentinel5p  | s5p_co_granule_available     | 06-01 18Z 1, 06-02 18Z 1, 06-03 18Z 1                                                                                                                                                        |
| sentinel5p  | s5p_hcho_column              | 06-01 18Z 0.0001645, 06-02 18Z 0.0001692                                                                                                                                                     |
| sentinel5p  | s5p_hcho_granule_available   | 06-01 18Z 1, 06-02 18Z 1, 06-03 18Z 1                                                                                                                                                        |
| sentinel5p  | s5p_no2_column               | 06-01 18Z 4.08e-05, 06-02 18Z 4.07e-05                                                                                                                                                       |
| sentinel5p  | s5p_no2_granule_available    | 06-01 18Z 1, 06-02 18Z 1, 06-03 18Z 1                                                                                                                                                        |
| sentinel5p  | s5p_o3_granule_available     | 06-01 18Z 1, 06-02 18Z 1, 06-03 18Z 1                                                                                                                                                        |
| sentinel5p  | s5p_so2_column               | 06-01 18Z 1.316e-05, 06-02 18Z -8.155e-06                                                                                                                                                    |
| sentinel5p  | s5p_so2_granule_available    | 06-01 18Z 1, 06-02 18Z 1, 06-03 18Z 1                                                                                                                                                        |
| tceq        | co                           | 06-01 00Z 0.2417, 06Z 0.2833, 12Z 0.2529, 18Z 0.2278, 06-02 00Z 0.2, 06Z 0.2333, 12Z 0.2056, 18Z 0.1667, 06-03 00Z 0.1389, 06Z 0.1222, 12Z 0.1667, 18Z 0.2, 06-04 00Z 0.2111                 |
| tceq        | no2                          | 06-01 00Z 3.283, 06Z 5.427, 12Z 6.123, 18Z 5.614, 06-02 00Z 4.532, 06Z 5.082, 12Z 7.521, 18Z 6.109, 06-03 00Z 5.897, 06Z 6.299, 12Z 8.659, 18Z 12.24, 06-04 00Z 11.52                        |
| tceq        | so2                          | 06-01 00Z -0.1583, 06Z -0.1778, 12Z 0.1889, 18Z 0.08333, 06-02 00Z -0.1444, 06Z -0.1294, 12Z -0.01176, 18Z 0.1556, 06-03 00Z -0.01667, 06Z -0.02778, 12Z 0.55, 18Z 0.1313, 06-04 00Z -0.2222 |

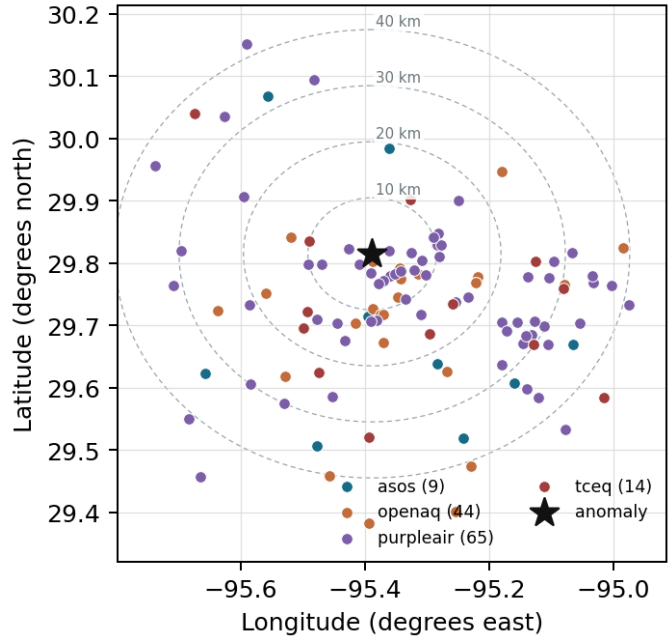
## Per-site averages

Each metric averaged at each site, with that site's distance from the event in brackets.

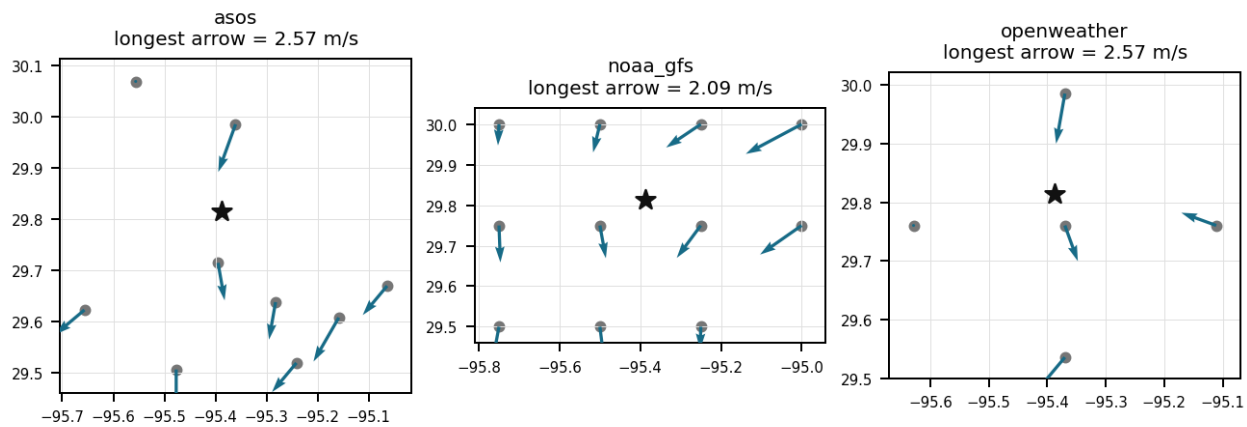
| Source      | Metric             | Values                                                                                                                                                                                                                                                                                                          |
|-------------|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| asos        | humidity           | MCJ (11.201 km) 80.81, IAH (19.067 km) 82.61, HOU (22.155 km) 79.17, EFD (31.938 km) 82.34, DWH (32.527 km) 85.73, SGR (33.627 km) 82.35, T41 (35.174 km) 82.52, AXH (35.363 km) 77.42, +1 more                                                                                                                 |
| asos        | temperature        | MCJ (11.201 km) 27, IAH (19.067 km) 26.62, HOU (22.155 km) 27.25, EFD (31.938 km) 27.09, DWH (32.527 km) 25.59, SGR (33.627 km) 27.01, T41 (35.174 km) 26.88, AXH (35.363 km) 26.97, +1 more                                                                                                                    |
| asos        | wind_direction     | MCJ (11.201 km) 107.4, IAH (19.067 km) 99.44, HOU (22.155 km) 106.1, EFD (31.938 km) 93.44, DWH (32.527 km) 70.83, SGR (33.627 km) 111.5, T41 (35.174 km) 93.75, AXH (35.363 km) 72.08, +1 more                                                                                                                 |
| asos        | wind_speed         | MCJ (11.201 km) 3.586, IAH (19.067 km) 2.522, HOU (22.155 km) 2.922, EFD (31.938 km) 2.452, DWH (32.527 km) 1.543, SGR (33.627 km) 2.648, T41 (35.174 km) 2.751, AXH (35.363 km) 1.693, +1 more                                                                                                                 |
| noaa_gfs    | gh_500             | gfs:29.75,-95.50 (12.999 km) 5885, gfs:29.75,-95.25 (15.102 km) 5885, gfs:30.00,-95.50 (23.292 km) 5884, gfs:30.00,-95.25 (24.525 km) 5884, gfs:29.75,-95.75 (35.695 km) 5885, gfs:29.50,-95.50 (36.619 km) 5886, gfs:29.50,-95.25 (37.419 km) 5886, gfs:29.75,-95.00 (38.097 km) 5885, +3 more                 |
| noaa_gfs    | pbl_height         | gfs:29.75,-95.50 (12.999 km) 665.6, gfs:29.75,-95.25 (15.102 km) 618.5, gfs:30.00,-95.50 (23.292 km) 691.2, gfs:30.00,-95.25 (24.525 km) 658.7, gfs:29.75,-95.75 (35.695 km) 501.5, gfs:29.50,-95.50 (36.619 km) 508.6, gfs:29.50,-95.25 (37.419 km) 454.1, gfs:29.75,-95.00 (38.097 km) 440.4, +3 more         |
| noaa_gfs    | precipitable_water | gfs:29.75,-95.50 (12.999 km) 49.17, gfs:29.75,-95.25 (15.102 km) 50, gfs:30.00,-95.50 (23.292 km) 48.33, gfs:30.00,-95.25 (24.525 km) 49.07, gfs:29.75,-95.75 (35.695 km) 48.78, gfs:29.50,-95.50 (36.619 km) 50.03, gfs:29.50,-95.25 (37.419 km) 50.69, gfs:29.75,-95.00 (38.097 km) 50.93, +3 more            |
| noaa_gfs    | surface_pressure   | gfs:29.75,-95.50 (12.999 km) 1011, gfs:29.75,-95.25 (15.102 km) 1013, gfs:30.00,-95.50 (23.292 km) 1010, gfs:30.00,-95.25 (24.525 km) 1011, gfs:29.75,-95.75 (35.695 km) 1010, gfs:29.50,-95.50 (36.619 km) 1012, gfs:29.50,-95.25 (37.419 km) 1013, gfs:29.75,-95.00 (38.097 km) 1013, +3 more                 |
| noaa_gfs    | t_850              | gfs:29.75,-95.50 (12.999 km) 17.81, gfs:29.75,-95.25 (15.102 km) 17.87, gfs:30.00,-95.50 (23.292 km) 17.84, gfs:30.00,-95.25 (24.525 km) 17.79, gfs:29.75,-95.75 (35.695 km) 17.75, gfs:29.50,-95.50 (36.619 km) 17.82, gfs:29.50,-95.25 (37.419 km) 17.85, gfs:29.75,-95.00 (38.097 km) 17.83, +3 more         |
| noaa_gfs    | u_10m              | gfs:29.75,-95.50 (12.999 km) -1.508, gfs:29.75,-95.25 (15.102 km) -1.711, gfs:30.00,-95.50 (23.292 km) -1.415, gfs:30.00,-95.25 (24.525 km) -1.793, gfs:29.75,-95.75 (35.695 km) -1.415, gfs:29.50,-95.50 (36.619 km) -1.912, gfs:29.50,-95.25 (37.419 km) -1.925, gfs:29.75,-95.00 (38.097 km) -1.695, +3 more |
| noaa_gfs    | v_10m              | gfs:29.75,-95.50 (12.999 km) 0.5034, gfs:29.75,-95.25 (15.102 km) 0.3951, gfs:30.00,-95.50 (23.292 km) 0.3117, gfs:30.00,-95.25 (24.525 km) 0.2351, gfs:29.75,-95.75 (35.695 km) 0.5134, gfs:29.50,-95.50 (36.619 km) 0.1934, gfs:29.50,-95.25 (37.419 km) 0.3809, gfs:29.75,-95.00 (38.097 km) 0.7526, +3 more |
| openaq      | ozone              | 319 (10.034 km) 0.02019, 2046 (10.097 km) 0.01817, 310 (11.311 km) 0.02499, 3378 (15.436 km) 0.01757, 325 (16.839 km) 0.01803, 2039 (16.93 km) 0.01938, 311 (17.034 km) 0.02164, 312 (22.784 km) 0.02, +6 more                                                                                                  |
| openaq      | pm10               | 13380082 (15.436 km) 30.54, 271 (29.738 km) 20.7                                                                                                                                                                                                                                                                |
| openaq      | pm25               | 2071327 (0.0 km) 9.156, 10107877 (1.388 km) 7.53, 8967193 (4.94 km) 6.703, 15596557 (6.196 km) 5.826, 8967217 (7.933 km) 5.499, 15666344 (8.761 km) 3.178, 12286630 (9.785 km) 2.814, 2071331 (10.097 km) 10.55, +20 more                                                                                       |
| openweather | cloud_cover        | grid:center (6.262 km) 59.44, grid:north (19.034 km) 61.65, grid:west (24.006 km) 53.32, grid:east (27.362 km) 55.31, grid:south (31.042 km) 47.81                                                                                                                                                              |
| openweather | humidity           | grid:center (6.262 km) 80.49, grid:north (19.034 km) 81.67, grid:west (24.006 km) 80.58, grid:east (27.362 km) 85.18, grid:south (31.042 km) 79.5                                                                                                                                                               |
| openweather | precipitation      | grid:center (6.262 km) 0.2676, grid:north (19.034 km) 0.3278, grid:west (24.006 km) 0.3024, grid:east (27.362 km) 0.3336, grid:south (31.042 km) 0.2425                                                                                                                                                         |
| openweather | pressure           | grid:center (6.262 km) 1014, grid:north (19.034 km) 1014, grid:west (24.006 km) 1014, grid:east (27.362 km) 1014, grid:south (31.042 km) 1014                                                                                                                                                                   |
| openweather | temperature        | grid:center (6.262 km) 27.61, grid:north (19.034 km) 27.11, grid:west (24.006 km) 27.32, grid:east (27.362 km) 27.23, grid:south (31.042 km) 27.28                                                                                                                                                              |
| openweather | wind_direction     | grid:center (6.262 km) 115.2, grid:north (19.034 km) 108.8, grid:west (24.006 km) 140.1, grid:east (27.362 km) 85.89, grid:south (31.042 km) 101.6                                                                                                                                                              |
| openweather | wind_speed         | grid:center (6.262 km) 2.701, grid:north (19.034 km) 2.213, grid:west (24.006 km) 1.924, grid:east (27.362 km) 2.4, grid:south (31.042 km) 2.179                                                                                                                                                                |

| Source    | Metric | Values                                                                                                                                                                                                                                      |
|-----------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| purpleair | pm25   | 246011 (2.65 km) 5.599, 223813 (2.654 km) 6.796, 302037 (3.453 km) 7.445, 155367 (3.78 km) 7.463, 127451 (4.849 km) 1.882, 186303 (5.082 km) 7.475, 27821 (5.158 km) 0.3384, 296101 (5.395 km) 7.256, +57 more                              |
| tceq      | co     | 482011052 (0.02 km) 0.2863, 482011035 (15.439 km) 0.1803, 482011039 (29.754 km) 0.1397                                                                                                                                                      |
| tceq      | no2    | 482011052 (0.02 km) 9.459, 482010047 (10.026 km) 8.458, 482010024 (11.303 km) 0.4941, 482011066 (14.47 km) 8.992, 482011035 (15.439 km) 8.785, 482010416 (16.848 km) 8.537, 482010055 (17.04 km) 6.203, 482010026 (25.33 km) 9.124, +5 more |
| tceq      | so2    | 482011035 (15.439 km) 0.01507, 482010051 (22.781 km) -0.2186, 482011039 (29.754 km) 0.2917                                                                                                                                                  |

Ground observation locations in the stored 72-hour context  
dashed rings are great-circle distance from the anomaly



Event-nearest wind vectors from stored values (arrow points downwind)



## Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). The labeling guide that comes with this packet has the definitions and marking rules. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

### Claim 1 of 36

Shallow boundary layer height under 50 meters combined with low surface wind speeds trapped localized emissions near the ground surface.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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### Claim 2 of 36

Regional wildfire smoke or a sulfur-rich industrial emission plume is weakened as an explanation because ozone, CO, NO<sub>2</sub>, and SO<sub>2</sub> near the anomaly were low at the event time and sentinel5p showed no elevated CO or SO<sub>2</sub> columns later that day over the area.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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### Claim 3 of 36

The air quality anomaly in Houston, Texas is caused by high levels of PM<sub>2.5</sub> particles.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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#### Claim 4 of 36

The anomaly is caused by a nearby industrial facility releasing excessive amounts of particulate matter into the atmosphere.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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#### Claim 5 of 36

A severe thunderstorm or tornado event has stirred up pollutants and particles from the ground, contributing to the air quality anomaly.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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#### Claim 6 of 36

NOAA GFS model simulations and ASOS surface observations showed a shallow planetary boundary layer height below 50 meters and light wind speeds near 1.5 m/s surrounding the anomaly time.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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#### Claim 7 of 36

On June 2, 2026, at 14:00 UTC, an OpenAQ monitor in Houston (29.815, -95.388) recorded a severe PM2.5 concentration of 34.5 ug/m3, well above the expected value of 10.75 ug/m3 with a z-score of 7.39.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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**Claim 8 of 36**

The anomaly is a result of a combination of factors including high temperatures, low humidity, and stagnant atmospheric conditions, which have led to an accumulation of pollutants in the area.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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**Claim 9 of 36**

Meteorological stagnation and moisture favored temporary accumulation or enhancement of particulate matter because winds near the anomaly were weak at about 1.5 m/s, humidity was high at about 84 to 86 percent, and gfs planetary boundary layer height was extremely low near 48 m around the event time.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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**Claim 10 of 36**

The anomaly occurred on June 2nd at approximately 14:00:00+00:00 (0.0 min away) and is located near sensor 246011 (2.65 km).

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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**Claim 11 of 36**

NO<sub>2</sub> levels were also elevated, with a mean value of 6.7047 ppb, suggesting a potential contribution from nearby industrial activities or vehicle emissions.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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**Claim 12 of 36**

The PM<sub>2.5</sub> levels exceeded the normal range, with a mean value of 6.437 ug/m<sup>3</sup>, indicating a significant increase in particulate matter.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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**Claim 13 of 36**

A localized primary PM<sub>2.5</sub> emission plume near the monitor is a leading hypothesis because the anomalous PM<sub>2.5</sub> value at the openaq site was not mirrored by the nearest purpleair sensor, indicating strong sub-kilometer spatial heterogeneity consistent with a nearby combustion or industrial particle source.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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**Claim 14 of 36**

The PM<sub>2.5</sub> levels are elevated at 4.6 ug/m<sup>3</sup>, which is higher than the mean of 6.437 ug/m<sup>3</sup>.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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**Claim 15 of 36**

The nearest PurpleAir PM2.5 monitor located 2.65 km away recorded a concentration of 4.6 ug/m3 at the time of the anomaly on June 2, 2026, at 14:00 UTC.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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**Claim 16 of 36**

Meteorological conditions near central Houston on 2026-06-02 14:00 UTC were favorable for short-term pollutant buildup because local winds were light at about 1.5 m/s, relative humidity was high at about 84 to 86 percent, and gfs boundary-layer height was extremely low at about 48 m near 12:00 UTC.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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**Claim 17 of 36**

Mechanical resuspension of fine dust or mixed road dust is a plausible hypothesis because PM10 was elevated at the same time as the PM2.5 spike while local winds were light, a pattern consistent with a local coarse-plus-fine particle source affecting one part of the urban area.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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**Claim 18 of 36**

High relative humidity levels exceeding 80% measured by ASOS and OpenWeather during the morning of June 2, 2026, enhanced hygroscopic growth of fine particulate matter.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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**Claim 19 of 36**

High relative humidity exceeding 80 percent promoted hygroscopic growth of fine aerosol particles and localized particle accumulation.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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**Claim 20 of 36**

The air quality anomaly in Houston, Texas is caused by high levels of NO2 particles.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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**Claim 21 of 36**

At 2026-06-02T14:00:00+00:00 near the anomaly location, openaq ozone was 0.009 ppm, tceq CO was 0.1 ppm, tceq NO2 was 2.0 ppb, and openaq PM10 at the nearest reporting site was 51.0 ug/m3.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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**Claim 22 of 36**

Meteorological trapping and hygroscopic enhancement likely amplified the measured PM<sub>2.5</sub> concentration because humidity was high and modeled planetary boundary layer height was extremely low near the event time, favoring poor dispersion and water uptake by aerosol.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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**Claim 23 of 36**

A localized primary particle plume or near-source dust/resuspension event is supported because openaq measured a sharp PM<sub>2.5</sub> spike to 34.5 ug/m<sup>3</sup> at the anomaly location while nearby purpleair remained much lower at 4.6 ug/m<sup>3</sup> within 2.65 km, indicating strong spatial heterogeneity consistent with a very local source or instrument-near plume rather than a uniform citywide event.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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**Claim 24 of 36**

A localized industrial or combustion point-source emission near the OpenAQ monitoring site caused the sharp PM<sub>2.5</sub> spike, which was not observed by adjacent PurpleAir sensors.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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### Claim 25 of 36

The air quality anomaly is related to PM2.5 levels.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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### Claim 26 of 36

The air quality anomaly in Houston, Texas is caused by high levels of SO2 particles.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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### Claim 27 of 36

The openaq PM2.5 anomaly at 29.815, -95.388 on 2026-06-02 14:00 UTC is most consistent with a localized primary particle source because openaq measured 34.5 ug/m3 at the anomaly site while nearby gaseous co-pollutants remained low at the same time, including CO of 0.1 ppm, NO2 of 2.0 ppb, ozone of 0.009 ppm, and SO2 near zero.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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### Claim 28 of 36

The air quality anomaly in Houston, Texas on June 2nd was caused by a combination of factors including high temperatures, low wind speeds, and nearby industrial activities.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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### Claim 29 of 36

On June 2, 2026, at 14:00 UTC, an OpenAQ monitoring station in Houston recorded an elevated PM2.5 concentration of 34.5  $\mu\text{g}/\text{m}^3$  against an expected baseline of 10.75  $\mu\text{g}/\text{m}^3$ .

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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### Claim 30 of 36

High surface relative humidity above 83% reported by ASOS and OpenWeather near the time of the anomaly supports aerosol hygroscopic growth as a contributing factor to elevated PM2.5 mass concentrations.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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### Claim 31 of 36

Low PM2.5 readings at surrounding PurpleAir and OpenAQ sensors during the anomaly window indicate that the PM2.5 elevation was isolated to the immediate area of the reporting monitor rather than part of a regional plume.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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### Claim 32 of 36

An OpenAQ PM10 sensor located 15.436 km from the anomaly recorded an elevated PM10 concentration of 51.0  $\mu\text{g}/\text{m}^3$  at 2026-06-02T14:00:00+00:00.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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### Claim 33 of 36

The openaq PM2.5 monitor at 29.815, -95.388 in Houston measured 34.5 ug/m3 at 2026-06-02T14:00:00+00:00.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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### Claim 34 of 36

The openaq PM2.5 anomaly at 29.815, -95.388 in Houston had an expected value of 10.75 ug/m3 and a z-score of 7.3937237743124316 at 2026-06-02T14:00:00+00:00, which was classified as severe.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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### Claim 35 of 36

Model output from GFS and surface observations from ASOS indicate a shallow planetary boundary layer below 50 meters and light surface winds under 2 m/s around 14:00 UTC, supporting surface boundary layer trapping of fine particulates.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:

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Claim 36 of 36

Regional transported smoke or a large sulfur-rich industrial plume is less supported because sentinel5p showed no contemporaneous enhancement in CO or SO2 columns over the area and purpleair nearby remained much lower than the anomaly site at about 4.6 ug/m3.

|                   |                            |                            |                            |
|-------------------|----------------------------|----------------------------|----------------------------|
| Mark exactly one: | <input type="checkbox"/> V | <input type="checkbox"/> I | <input type="checkbox"/> U |
|-------------------|----------------------------|----------------------------|----------------------------|

Note:



**Optional: most likely cause**

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.

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